Claas Beger

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Education

Cornell University — *Aug 2024 – May 2025*
MEng in Computer Science — *GPA: 4.21/4.3*

Hong Kong University of Science and Technology — *Jan 2022 – May 2022*
Exchange - Computer Science — *GPA: 4.1/4.3*

Technical University of Munich — *Oct 2019 – Mar 2024*
BS in Information Systems — *GPA: 3.7/4.0*

Experience

Research Scientist — *New York, New York*
Stealth Startup — *July 2026 – Present*

- Developing label-free distillation methods for training models without human-annotated supervision.
- Building evaluations that characterize jagged intelligence, i.e. uneven capability profiles across tasks.

Graduate Fellow — *Santa Fe, New Mexico*
Santa Fe Institute — *June 2025 – June 2026*

- Researching multimodal abstract reasoning in AI models under Professor Melanie Mitchell.
- Designing evaluation benchmarks and analysis pipelines for scientific reasoning with Sandia National Laboratories.

Graduate Teaching and Research Specialist — *Ithaca, New York*
Cornell University — *Jan 2025 – May 2025*

- Co-designed assignments, projects, and exams for *Introduction to Machine Learning*.
- Graded and provided feedback on students' work; held office hours for Q&A.

Responsible AI Engineering Intern — *Munich, Bavaria*
PwC — *June 2022 – Jan 2023*

- Integrated SHAP and LIME into an explainable AI interface for TensorFlow, PyTorch, and sklearn models.
- Co-authored and presented a white paper at Bitkom Big Data & AI Summit Berlin 2022.

Financial Technologies SWE Intern — *Frankfurt, Hessia*
Deutsche Bank — *Oct 2021 – Jan 2022*

- Developed microservices in Python and Java (Spring Boot) on Red Hat OpenShift.
- Managed database migration and analysis workflows using Flywaydb, JDBC, JPA, and PostgreSQL.
- Collaborated with teams to integrate services into trading and risk management systems.

Research Assistant — *Munich, Bavaria*
Institute of Computational Biology Munich — *Sep 2020 – Jun 2021*

- Built neural networks for classification and clustering of 20+ million single-cell samples using TensorFlow and Keras.
- Developed autoencoders to generate synthetic diabetic cell samples to augment datasets.
- Processed and visualized biological data in collaboration with computational biologists.

Teaching Assistant — *Munich, Bavaria*
TU Munich; Chairs of Information Systems and Applied Software Engineering — *Oct 2020 – Sep 2021*

- Led weekly tutorials on software engineering, design patterns, model-based engineering, and information systems.
- Developed exam questions, assisted with grading and lecture preparation, and held office hours for student support.

Technologies

Technologies: Python, C++, Java, Swift, vLLM, PyTorch, TensorFlow, peft, DeepSpeed, Pandas, SpringBoot

Skills: Machine Learning, NLP, Computer Vision, Software Engineering, Neuromorphic Algorithms

Publications

To Appear

PotARCin: Multi-Dimensional Evaluation of Skill Acquisition in Abstract Reasoning Tasks. ***Claas Beger***, Ryan Yi, Melanie Mitchell. To appear in Advances in Neural Information Processing Systems (NeurIPS 2026). arXiv

Distinguishing Performance From Competence in Evaluations of Humanlike Abstract Reasoning. ***Claas Beger***, Ryan Yi, Shuhao Fu, Arseny Moskvichev, Sarah W. Tsai, Sivasankaran Rajamanickam, Melanie Mitchell. To appear in Advances in Neural Information Processing Systems (NeurIPS 2026). arXiv

Illusions of Abstraction in Contemporary AI. Melanie Mitchell, Jacob G. Foster, ***Claas Beger***, Ryan Yi, Shuhao Fu, Kaleda K. Denton, Alessandro Palmarini. To appear in Proceedings of the National Academy of Sciences (PNAS).

Published

π-CoT: Prolog-Initialized Chain-of-Thought Prompting for Multi-Hop Question-Answering. Chao Wan, Albert Gong, Mihir Mishra, Carl-Leander Henneking, ***Claas Beger***, Kilian Q. Weinberger. Transactions on Machine Learning Research (TMLR), 2026. Paper arXiv

OmniCode: A Benchmark for Evaluating Software Development Agents. Atharv Sonwane*, Eng-Shen Tu*, Wei-Chung Lu*, ***Claas Beger****, ..., Kevin Ellis, Saikat Dutta. Findings of the Association for Computational Linguistics: ACL 2026. Paper arXiv

Bongards at the Boundary of Perception and Reasoning: Programs or Language? Cassidy Langenfeld, ***Claas Beger***, Gloria Geng, Wasu Top Piriyakulkij, Keya Hu, Yewen Pu, Kevin Ellis. 48th Annual Meeting of the Cognitive Science Society (CogSci 2026). arXiv

Cognitive-Science–Inspired Evaluation of Large Language Models. Shuhao Fu, ***Claas Beger***, Ryan Yi, Melanie Mitchell, Michael C. Frank, Raphaël Millière, Robert Hawkins, Tomer D. Ullman. Symposium at the 48th Annual Meeting of the Cognitive Science Society (CogSci 2026). Paper

Investigating Abstraction Capabilities of the o3 Model Using Textual and Visual Modalities. ***Claas Beger***, Shuhao Fu, Ryan Yi, Arseny Moskvichev, Melanie Mitchell. NeurIPS 2025 Workshop on Multimodal Algorithmic Reasoning, Nov 2025. Paper • **Spotlight Presentation**

Refining Inverse Constitutional AI for Dataset Validation Under the EU AI Act. Carl-Leander Henneking*, ***Claas Beger****. Regulatable ML Workshop at NeurIPS 2025, Nov 2025. Paper

A Neuroscience-Inspired Dual-Process Model of Compositional Generalization. Alex Noviello*, ***Claas Beger****, Jacob Groner, Kevin Ellis, Weinan Sun. NeurIPS 2025 Workshop on Interpreting Cognition in Deep Learning Models, Nov 2025. Paper

Towards AI Collaborators: Exploring Goal, Value, and Role-Based Alignment in AI. ***Claas Beger***. 51st Annual Meeting of the Society for Philosophy and Psychology (SPP), Jun 2025. Poster • **Poster Award**

CoCoNUT: Structural Code Understanding Does Not Fall Out of a Tree. ***Claas Beger***, Saikat Dutta. LLM4Code Workshop at ICSE 2025, Jan 2025. Paper

Under Review

SpikeDecoder: Realizing the GPT Architecture with Spiking Neural Networks. ***Claas Beger***, Florian Walter, Alois Knoll. Under review, Jun 2026. arXiv

Citegeist: Automated Generation of Related Work Analysis on the arXiv Corpus. ***Claas Beger****, Carl-Leander Henneking*. Under review, Jun 2025. arXiv • **Statistics and Data Science Faculty Award**

** indicates equal contribution*

Awards and Recognitions

Diverse Intelligences Fellowship (6000 USD) *Feb 2026*
- Awarded by the Natural and Artificial Minds Initiative, Princeton University.

Poster Award, Society for Philosophy and Psychology (150 USD) *Jun 2025*
- Awarded for "Towards AI Collaborators: Exploring Goal, Value, and Role-Based Alignment in AI."

Department of Statistics and Data Science Faculty Award (1000 USD) *Apr 2025*
- Awarded by Cornell University, for the research project "Citegeist".

Dean's List, Hong Kong University of Science and Technology *Jun 2022*
- Top 5% of the undergraduate student body.